

LIFEWS AgriShine(TM) GrowMeal(TM)

SSS 1-3 RESEARCH & ENGINEERING JOURNAL

STUDENT EDITION

SYSTEM / RESEARCH SKETCH

Name: _____

Class: _____ School: _____

**FRAME -> DESIGN -> INSTRUMENT -> MODEL -> VALIDATE -> EVALUATE -> OPTIMIZE ->
RECOMMEND**

SCIENTIFIC / ENGINEERING RECORD STANDARD

Use dated, traceable records. Define variables and units before collecting data. Record unexpected values rather than deleting them. Distinguish observations from interpretations. Follow teacher-approved safety procedures and instrument instructions.

Record item	What I must document

MY INSTRUMENT & DATA-QUALITY REGISTER

Instrument / method	Variable	Unit / range	Calibration / quality check

Where relevant, record sample size, replicate number, treatment, control, timestamp, sensor ID, uncertainty, missing data and corrective action.

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 01 - Advanced Production Garden

QUESTION / OBJECTIVE: What factors explain production performance?

METHOD: Build a dataset of selected inputs and yield components.

PRIMARY METRIC: Yield components

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Analyze dominant contributors.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 02 - Advanced Production Garden

QUESTION / OBJECTIVE: What is harvest index?

METHOD: Measure economic and total relevant biomass where feasible.

PRIMARY METRIC: Harvest index

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate and interpret.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 03 - Advanced Production Garden

QUESTION / OBJECTIVE: How intensive is resource use?

METHOD: Quantify selected inputs per unit area.

PRIMARY METRIC: Input intensity

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare management scenarios.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 04 - Advanced Production Garden

QUESTION / OBJECTIVE: What is resource-use efficiency?

METHOD: Relate output to a defined nutrient, water or other input.

PRIMARY METRIC: Efficiency ratio

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret limitations.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 05 - Advanced Production Garden

QUESTION / OBJECTIVE: Which management strategy is preferable?

METHOD: Compare performance, cost and resource indicators.

PRIMARY METRIC: Multi-criteria score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend and justify.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

ADVANCED PRODUCTION GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 06 - Crop Trial Garden

QUESTION / OBJECTIVE: What treatment effect is being tested?

METHOD: Define treatments, blocks/replicates and response variables.

PRIMARY METRIC: Treatment effect

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Specify experimental logic.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 07 - Crop Trial Garden

QUESTION / OBJECTIVE: Does blocking reduce unwanted variation?

METHOD: Compare within- and between-block patterns.

PRIMARY METRIC: Block variation

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate layout.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 08 - Crop Trial Garden

QUESTION / OBJECTIVE: How variable are replicate results?

METHOD: Calculate mean and standard deviation.

PRIMARY METRIC: Mean, SD

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret spread.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 09 - Crop Trial Garden

QUESTION / OBJECTIVE: How large is the treatment effect?

METHOD: Compare treatment means relative to reference.

PRIMARY METRIC: Effect size / percent change

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret practical significance.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 10 - Crop Trial Garden

QUESTION / OBJECTIVE: What recommendation does the trial support?

METHOD: Integrate effect, variability and constraints.

PRIMARY METRIC: Evidence summary

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Write a qualified recommendation.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

CROP TRIAL GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 11 - Controlled Experiment Garden

QUESTION / OBJECTIVE: What causal relationship is proposed?

METHOD: Define independent, response and controlled variables.

PRIMARY METRIC: Causal structure

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State hypothesis.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 12 - Controlled Experiment Garden

QUESTION / OBJECTIVE: Do two factors interact?

METHOD: Use a simple factorial layout.

PRIMARY METRIC: Factor responses

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret interaction pattern.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 13 - Controlled Experiment Garden

QUESTION / OBJECTIVE: Is confounding present?

METHOD: Audit design and uncontrolled differences.

PRIMARY METRIC: Confounding risk

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain threat to inference.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 14 - Controlled Experiment Garden

QUESTION / OBJECTIVE: How repeatable is the measurement?

METHOD: Repeat instrument measurements under stable conditions.

PRIMARY METRIC: Repeatability metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess measurement quality.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 15 - Controlled Experiment Garden

QUESTION / OBJECTIVE: How sensitive is the conclusion?

METHOD: Recalculate under plausible assumptions or exclusions.

PRIMARY METRIC: Sensitivity result

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess robustness.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

CONTROLLED EXPERIMENT GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 16 - Irrigation Engineering Garden

QUESTION / OBJECTIVE: What flow rate must the system deliver?

METHOD: Use area, application requirement and operating time.

PRIMARY METRIC: Design flow rate

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate with units.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 17 - Irrigation Engineering Garden

QUESTION / OBJECTIVE: What head must the pump overcome?

METHOD: Measure/estimate elevation and approved losses.

PRIMARY METRIC: Total dynamic head

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State assumptions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 18 - Irrigation Engineering Garden

QUESTION / OBJECTIVE: How does pipe diameter affect loss?

METHOD: Compare teacher-provided or measured scenarios.

PRIMARY METRIC: Head-loss comparison

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret engineering trade-off.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 19 - Irrigation Engineering Garden

QUESTION / OBJECTIVE: How uniform are emitters?

METHOD: Measure discharge from a representative sample.

PRIMARY METRIC: Distribution uniformity

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate performance.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 20 - Irrigation Engineering Garden

QUESTION / OBJECTIVE: How efficient is pumping?

METHOD: Relate hydraulic output to measured/known input where safe.

PRIMARY METRIC: Pump efficiency

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify improvement opportunities.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

IRRIGATION ENGINEERING GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 21 - Water Productivity Garden

QUESTION / OBJECTIVE: What is the field water balance?

METHOD: Track measured/estimated inflows, outflows and storage change.

PRIMARY METRIC: Water balance

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Check closure and uncertainty.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 22 - Water Productivity Garden

QUESTION / OBJECTIVE: What is crop water use?

METHOD: Use teacher-provided ET data or approved estimation method.

PRIMARY METRIC: ET

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Document method.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 23 - Water Productivity Garden

QUESTION / OBJECTIVE: What is irrigation water productivity?

METHOD: Relate yield to irrigation input.

PRIMARY METRIC: IWP

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret context.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 24 - Water Productivity Garden

QUESTION / OBJECTIVE: What is ET water productivity?

METHOD: Relate yield to evapotranspiration.

PRIMARY METRIC: ETWP

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare with IWP.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 25 - Water Productivity Garden

QUESTION / OBJECTIVE: How should limited water be allocated?

METHOD: Compare scenarios under a fixed water constraint.

PRIMARY METRIC: Marginal/total productivity

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend allocation.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

WATER PRODUCTIVITY GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 26 - Advanced Soilless Garden

QUESTION / OBJECTIVE: How is the recirculating system structured?

METHOD: Map reservoir, dosing, delivery, root zone and return.

PRIMARY METRIC: System architecture

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify control points.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 27 - Advanced Soilless Garden

QUESTION / OBJECTIVE: How stable are pH and EC?

METHOD: Collect synchronized pH and EC time series.

PRIMARY METRIC: pH/EC dynamics

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret drift and coupling.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 28 - Advanced Soilless Garden

QUESTION / OBJECTIVE: How does source-water quality matter?

METHOD: Use teacher-provided alkalinity or source-water data.

PRIMARY METRIC: Water-quality indicator

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain management implications.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 29 - Advanced Soilless Garden

QUESTION / OBJECTIVE: What is hydraulic residence time?

METHOD: Use system volume and flow rate.

PRIMARY METRIC: Residence time

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate and interpret.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 30 - Advanced Soilless Garden

QUESTION / OBJECTIVE: What evidence supports a nutrient/root-zone diagnosis?

METHOD: Integrate solution, root and plant observations.

PRIMARY METRIC: Diagnostic evidence

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend corrective action.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

ADVANCED SOILLESS GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 31 - Sensor and Monitoring Garden

QUESTION / OBJECTIVE: How accurate and precise is the sensor?

METHOD: Compare repeated sensor readings with a reference.

PRIMARY METRIC: Bias and spread

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Separate accuracy from precision.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 32 - Sensor and Monitoring Garden

QUESTION / OBJECTIVE: What is the calibration relationship?

METHOD: Collect reference-sensor pairs across a safe range.

PRIMARY METRIC: Calibration curve

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Fit/interpret relationship.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 33 - Sensor and Monitoring Garden

QUESTION / OBJECTIVE: What is MAE?

METHOD: Use paired observed/reference data.

PRIMARY METRIC: MAE

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate and interpret.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 34 - Sensor and Monitoring Garden

QUESTION / OBJECTIVE: What is RMSE?

METHOD: Use paired observed/reference data.

PRIMARY METRIC: RMSE

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate and compare with MAE.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 35 - Sensor and Monitoring Garden

QUESTION / OBJECTIVE: How should a monitoring network be designed?

METHOD: Use spatial variability, purpose and constraints.

PRIMARY METRIC: Sensor number/placement

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Justify network design.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

SENSOR AND MONITORING GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 36 - Smart Irrigation Demonstrator

QUESTION / OBJECTIVE: What variables define the controller?

METHOD: Identify state, control, disturbance and constraint variables.

PRIMARY METRIC: Control architecture

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Map relationships.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 37 - Smart Irrigation Demonstrator

QUESTION / OBJECTIVE: How does threshold control behave?

METHOD: Run or simulate teacher-approved threshold logic.

PRIMARY METRIC: On/off response

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate cycling.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 38 - Smart Irrigation Demonstrator

QUESTION / OBJECTIVE: What does hysteresis change?

METHOD: Compare single-threshold and hysteresis logic.

PRIMARY METRIC: Switching frequency

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret stability.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 39 - Smart Irrigation Demonstrator

QUESTION / OBJECTIVE: How does feedback respond to disturbance?

METHOD: Introduce/simulate a safe input change.

PRIMARY METRIC: Control response

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess recovery.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 40 - Smart Irrigation Demonstrator

QUESTION / OBJECTIVE: Does adaptive control outperform a benchmark?

METHOD: Compare water use and target-state performance.

PRIMARY METRIC: Performance metrics

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate trade-offs.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

SMART IRRIGATION DEMONSTRATOR - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 41 - Research and Design Garden

QUESTION / OBJECTIVE: What problem should the design solve?

METHOD: Translate stakeholder need into a precise problem statement.

PRIMARY METRIC: Problem definition

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Check scope.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 42 - Research and Design Garden

QUESTION / OBJECTIVE: What are the design requirements?

METHOD: Convert needs into measurable requirements and constraints.

PRIMARY METRIC: Requirement set

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Check testability.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 43 - Research and Design Garden

QUESTION / OBJECTIVE: Which concept is preferable?

METHOD: Build a weighted decision matrix.

PRIMARY METRIC: Decision score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Conduct sensitivity check.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 44 - Research and Design Garden

QUESTION / OBJECTIVE: How should the prototype be tested?

METHOD: Define test conditions, metrics and acceptance criteria.

PRIMARY METRIC: Test plan

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess completeness.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 45 - Research and Design Garden

QUESTION / OBJECTIVE: Is the design technically and economically promising?

METHOD: Use selected performance and cost metrics.

PRIMARY METRIC: Techno-economic metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State uncertainty and recommendation.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

RESEARCH AND DESIGN GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 46 - Integrated Food Systems Laboratory

QUESTION / OBJECTIVE: What is the integrated system boundary?

METHOD: Map food, water, energy and supporting components.

PRIMARY METRIC: Boundary and interfaces

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain exclusions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 47 - Integrated Food Systems Laboratory

QUESTION / OBJECTIVE: Does the mass balance close?

METHOD: Account for measured/estimated inputs, outputs and storage.

PRIMARY METRIC: Mass-balance error

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Quantify closure.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 48 - Integrated Food Systems Laboratory

QUESTION / OBJECTIVE: How do supply and demand vary with time?

METHOD: Compare synchronized profiles.

PRIMARY METRIC: Mismatch metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify critical periods.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 49 - Integrated Food Systems Laboratory

QUESTION / OBJECTIVE: What cascade follows a component failure?

METHOD: Trace dependencies through the system.

PRIMARY METRIC: Cascade pathway

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify resilience intervention.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 50 - Integrated Food Systems Laboratory

QUESTION / OBJECTIVE: Which integrated strategy performs best?

METHOD: Run two or more scenarios against multiple objectives.

PRIMARY METRIC: Scenario metrics

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend a strategy and trade-offs.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INTEGRATED FOOD SYSTEMS LABORATORY - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

INDEPENDENT CAPSTONE INVESTIGATION

Research/design question: _____

Hypothesis or performance target: _____

Independent/design variable(s): _____

Response/performance variable(s): _____

Controls, constraints and assumptions: _____

Run / replicate	Input / treatment	Response	Unit / QC

ANALYSIS / MODEL / GRAPH

Evidence-based conclusion and recommendation: _____

MY GROWMEAL(TM) SSS 1-3 CERTIFICATE

This journal belongs to

who completed the GrowMeal(TM) investigation and research journal.

**FRAME -> DESIGN -> INSTRUMENT -> MODEL -> VALIDATE -> EVALUATE -> OPTIMIZE ->
RECOMMEND**

Teacher: _____ Date: _____